Module 15: Event-Driven Ansible

4 HOURS · 35% THEORY · 65% HANDS-ON

EDA Concepts

Understand event-driven automation

Rulebooks

Write EDA rulebooks

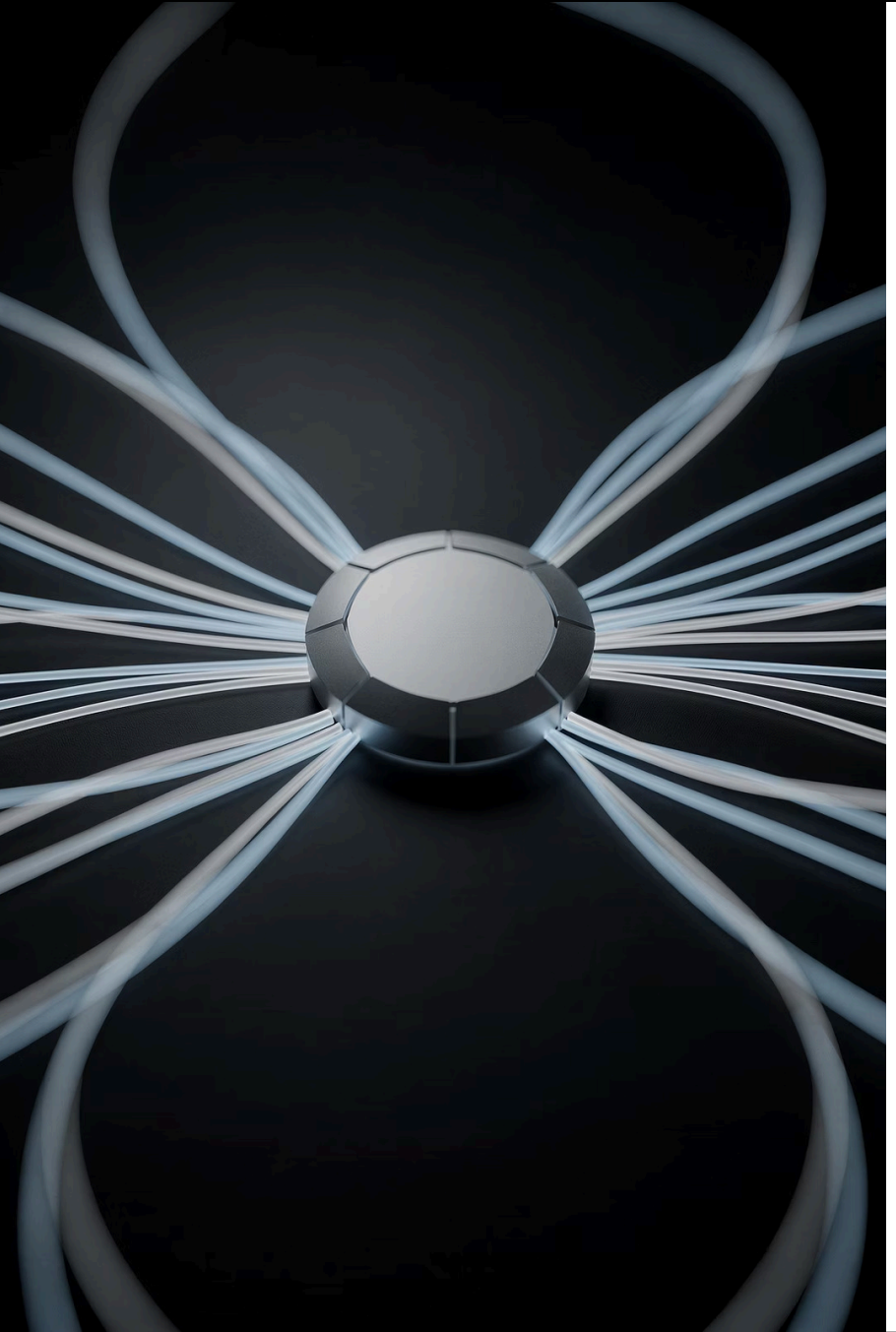
Integrations

Webhooks, Kafka, alerts

Deploy & Monitor

Active rulebooks in production

Event-Driven Ansible allows automation to **react automatically to events** instead of waiting for humans to run playbooks.



15.1 Introduction to Event-Driven Automation

What is Event-Driven Ansible (EDA)?

EDA allows automation to **react automatically when something happens** — no manual intervention needed.

EDA acts like an **automation event processor**.

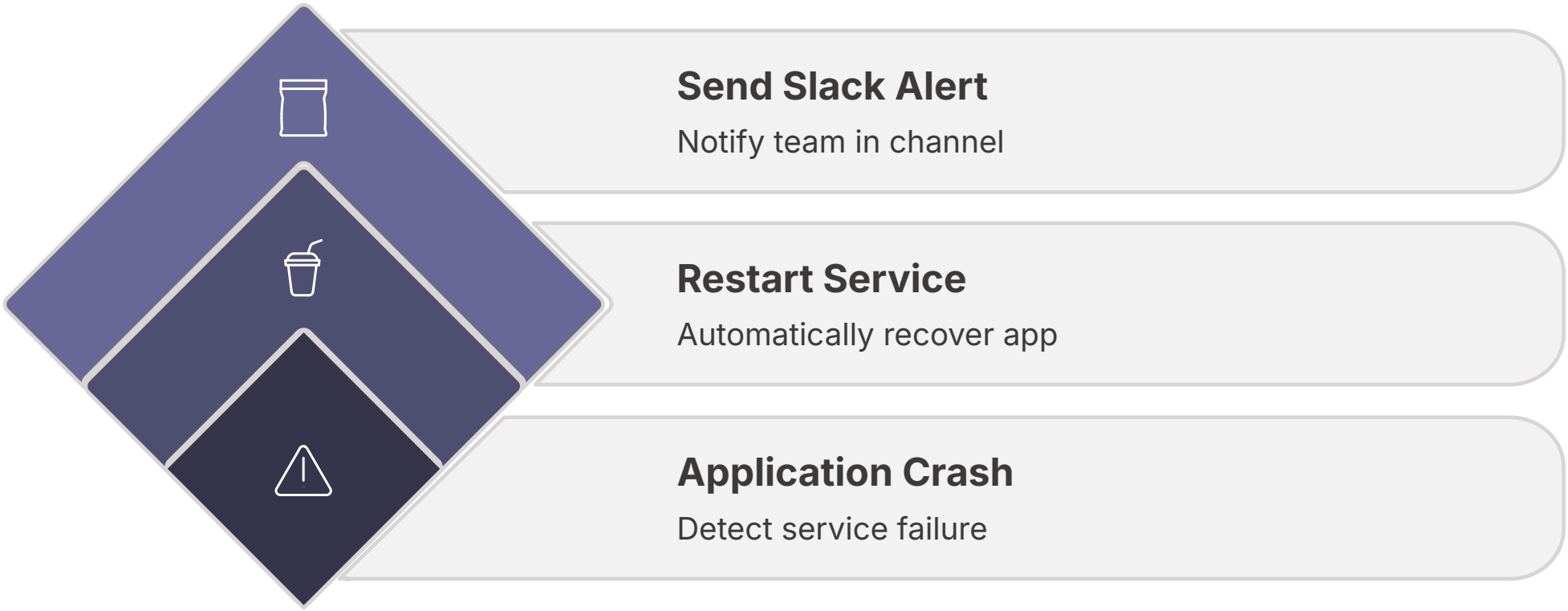
Event

Disk usage exceeds 90%

Response

Run cleanup playbook automatically

Simple Example



When an application crashes, EDA automatically handles both recovery and notification — no human needed.

Why Event-Driven Automation Matters

Traditional

Admin notices problem → Admin runs automation → Issue fixed

Event-Driven

Monitoring detects problem → Event sent to EDA → Automation triggers → Problem fixed immediately

This dramatically reduces **mean time to recovery (MTTR)**.

Traditional vs Event-Driven Automation

Traditional Automation — Pull Model

How it works

Automation runs on a **schedule or manually**.

Example: Run patch playbook every night

Manual or Scheduled

Execution is time-based, not event-based

Reactive After Detection

Response only after a human notices

Human Intervention Required

Someone must initiate the process

Event-Driven Automation — Push Model

How it works

Automation triggers **automatically based on events**.

If server CPU > 90% → run scaling playbook

Reactive Automation

Responds to conditions as they occur

Near Real-Time Response

Seconds, not minutes or hours

No Human Intervention

Fully autonomous execution

Side-by-Side Comparison

Feature	Traditional	Event-Driven
Execution	Manual / Scheduled	Automatic
Trigger	Human	Events
Response Time	Minutes / Hours	Seconds

Use Cases for Reactive Automation

Event-Driven automation is essential when **systems must react immediately**.

Infrastructure Auto-Remediation

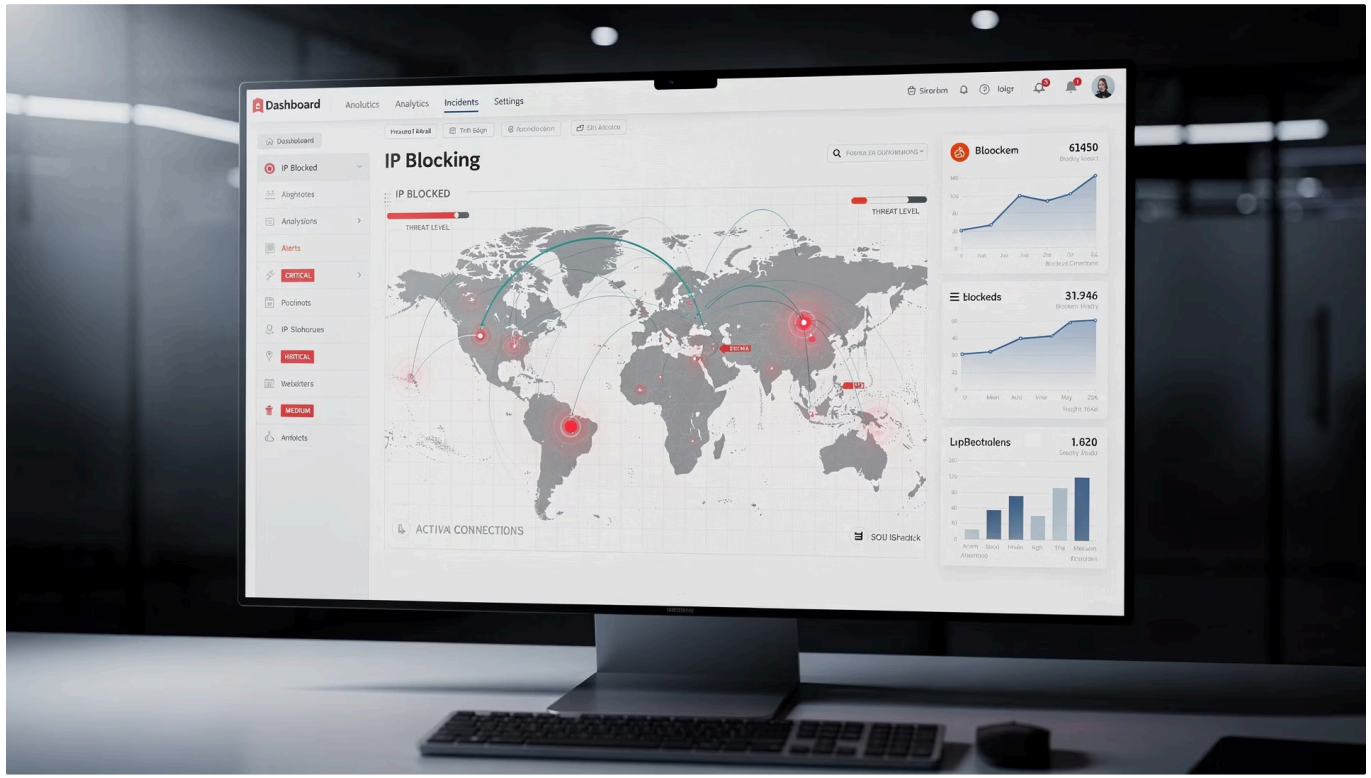
Trigger: Disk full alert

Delete Temp Files

Extend Storage

Notify Admin

Security Response



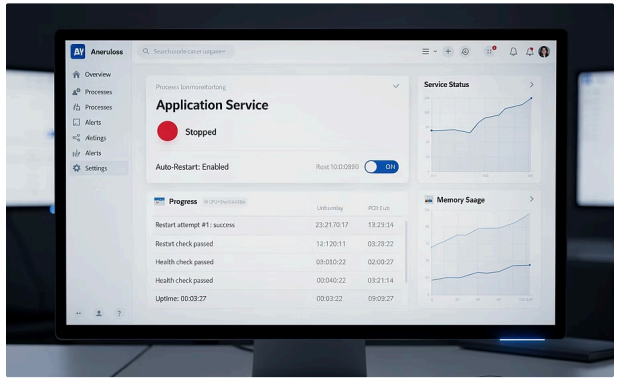
Trigger: Unauthorized login attempt

Block IP

Disable Account

Notify Security Team

Application Monitoring & Cloud Auto-Scaling



App Monitoring

Trigger: Application process stopped

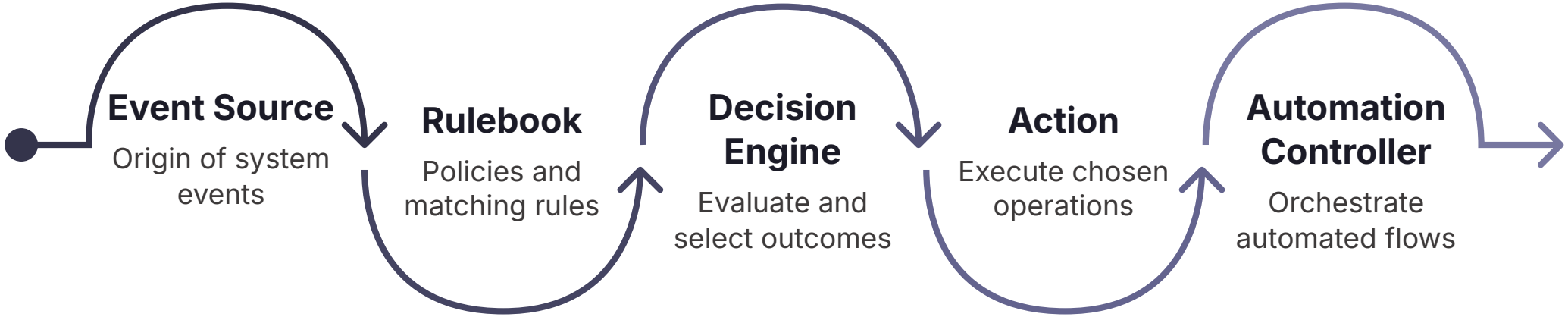
- Restart service
- Open incident ticket

Cloud Auto-Scaling

Trigger: High CPU usage

- Add additional compute nodes

EDA Architecture and Components



EDA introduces a layered architecture where each component plays a distinct role in processing and responding to events.

Core Components

Event Source

Generates events from external systems

Rulebook

Defines automation logic and conditions

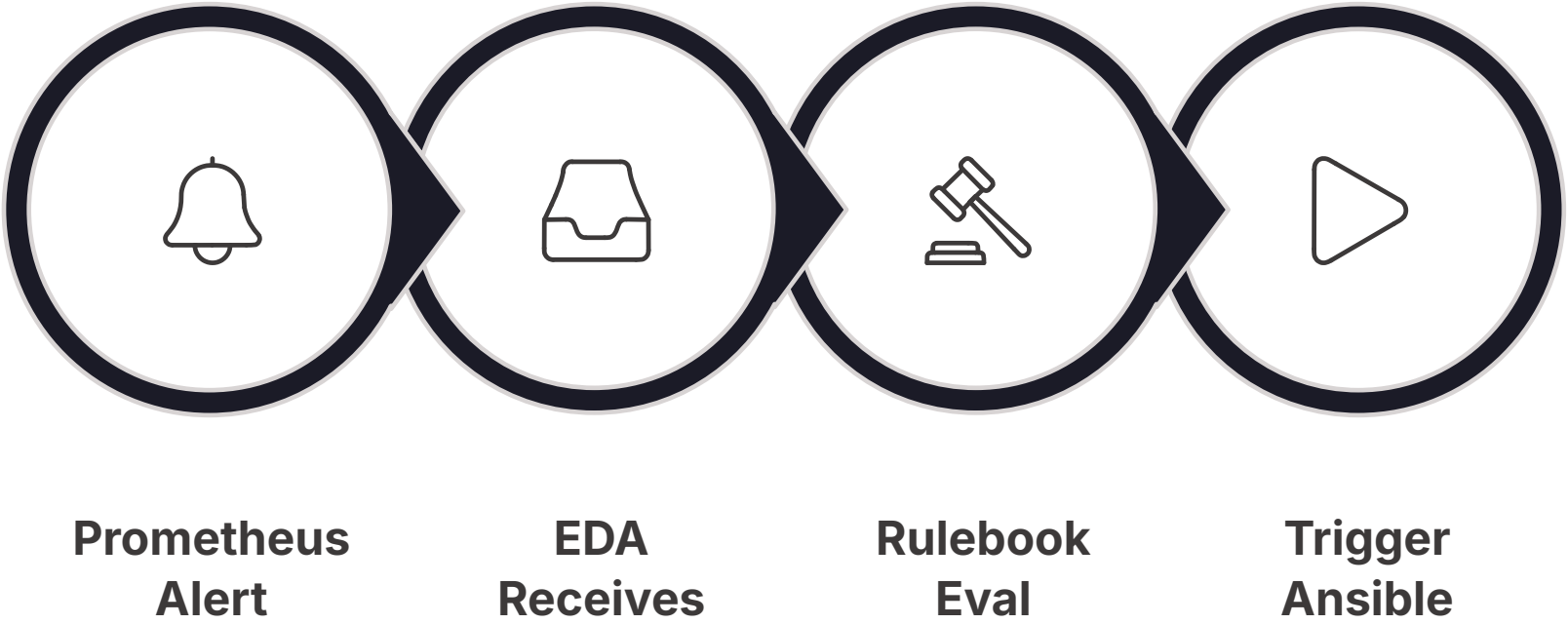
Decision Engine

Evaluates conditions against incoming events

Action

Executes the appropriate automation

Example Workflow



A Prometheus alert flows through EDA's decision engine, which evaluates the rulebook and triggers the appropriate Ansible playbook automatically.

When to Use EDA

Use EDA when:

→ Systems require **immediate response**

→ Automation depends on **external events**

→ Infrastructure is **dynamic and scalable**

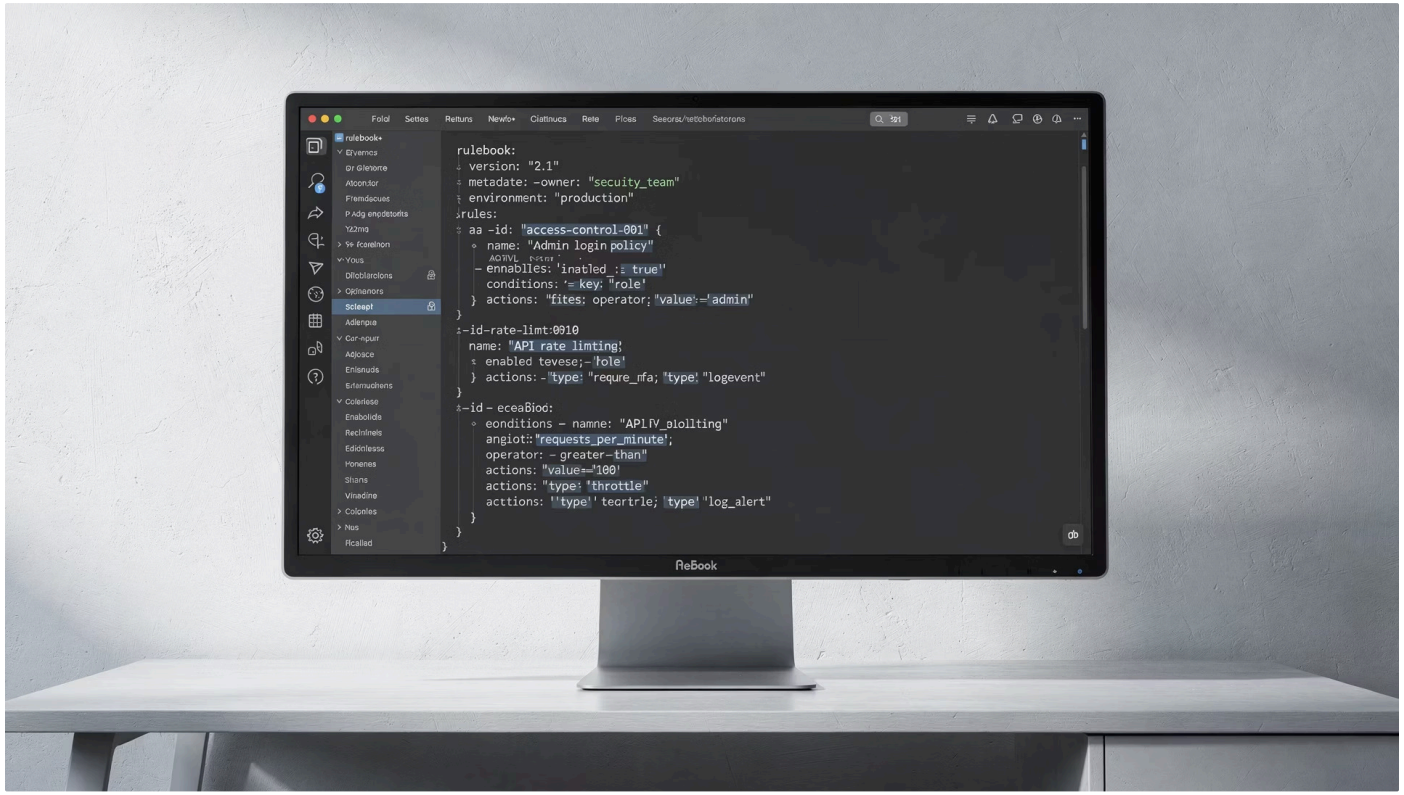
Common Examples

- Monitoring alerts
- Security incidents
- Cloud infrastructure changes
- CI/CD pipelines

EDA is **not always needed** — if automation runs on schedule, normal playbooks are sufficient.

15.2 Understanding Rulebooks

What is a Rulebook?



A **rulebook** is the core configuration file for Event-Driven Ansible — similar to playbooks but designed for events.

Event Sources

Where events come from

Conditions

When to act

Actions

What to execute